| NOT MEASUREMENT |
| SENSITIVE |

MIL-D-28594B
25 March 1989
SUPERSEDING
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MILITARY SPECIFICATION

DEHUMIDIFIER, ROTARY, DRY-DESICCANT TYPE, AUTOMATIC, CONTINUOUS DUTY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for dry desiccant, noncycling, rotary dehumidifiers.

1.2 Classification. The dehumidifier will be equipped with one of the following types of reactivation heaters, as specified (see 6.2):

Type I - Electric.
Type II - Gas.
Type III - Steam.

2. APPLICATION DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4440

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SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-P-40 - Packaging and Packing of Hand Tools.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose (Metric).
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-E-17555 - Electronic and Electrical Equipment Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-B-22191 - Barrier Materials, Transparent, Flexible, Heat Sealable.
MIL-H-22577 - Heating Elements, Electrical: Cartridge, Strip and Tubular Type.
MIL-B-26195 - Boxes, Wood-Cleated, Skidded, Load-Bearing Base.
MIL-P-46093 - Primer Coating, Synthetic (For Brake Drums).
MIL-C-52950 - Crates, Wood, Open and Covered.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing: Appropriate Test Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA AB1 - Molded Case Circuit Breakers. (DoD adopted)
NEMA ICS2 - Standards for Industrial Control Devices, Controllers, and Assemblies. (DoD adopted)
NEMA ICS6 - Enclosure for Industrial Controls and Systems. (DoD adopted)
NEMA MG1 - Standard for Motors and Generators. (DoD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code. (DoD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Lubrication Fittings. (DoD adopted)

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 489 - Model-Case Circuit Breakers and Circuit Breaker Enclosures. (DoD adopted)

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.2.1.

3.2 Standard commercial product. The dehumidifier shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product, shall be included in the dehumidifier being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs or brochures and represents the latest production model.

3.2.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.4 Design. The dehumidifier shall be a rotary, dry-desiccant type that automatically reactivates the desiccant, provides continuous operation, and is noncycling. It shall remove not less than 500 pounds (lb) of water per day when the entering air has a relative humidity of 35 percent and a dry-bulb temperature of 70 degrees Fahrenheit (^{o}F). The unit shall not contain a coolant. Humid air shall be continuously drawn into the unit, passed through the dry-desiccant, and then discharged into the space being dehumidified. Each unit, including all components, shall be mounted on a common base and shall be equipped with a lifting eye at each of the four upper corners or with four structural members suitable for the attachment of hoisting slings. It shall conform to the applicable safety requirements specified herein. Parts so located as to be hazardous to operating personnel and all rotating or reciprocating parts shall be fully enclosed or properly guarded. Protection to personnel shall be provided in accordance with NFPA No. 70. Screw-threads and pipe-threads shall conform to the requirements of FED-STD-H28.

3.5 Performance.

3.5.1 Capacity. The dehumidifier shall remove water under any ambient temperature between 0°F and 120°F at the rate indicated in table I (see 4.4.1).

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TABLE I. Water removal rate.

Entering air conditions		Minimum water removal rate, lb/24 hours
Dry bulb temperature °F	Relatively humidity percent	
70	35	500
55	35	296
40	35	164
30	35	114

3.5.2 Energy consumption. The reactivation energy required to remove each pound of water from the space being dehumidified shall not exceed the values indicated in table II (see 4.4.2). Total energy shall consist of the energy necessary to operate the heater, and when used on gas burners, the pre-mix blower as well as other required auxiliary components. Energy required to operate the reactivation blower shall not exceed 1.0 kilowatt.

TABLE II. Reactivation energy.

Entering air conditions		Energy per pound of water removed maximum		
		Type I	Type II	Type III
Dry bulb temperature °F	Relative humidity percent	Electric kWh	Gas Btu/h	Steam lb/h at 100 psig
70	35	0.9	4000	3.5
40	35	2.2	7300	6.3

3.6 Construction. The dehumidifier shall consist of a desiccant bed or contractor, a conditioned air fan and motor, a reactivation air fan and motor, a reactivation heater, air filters, and a drive system. They shall be mounted in controls, and thermal insulation. The dehumidifier shall be furnished complete for installation and, when connected to the required power source shall produce dry air of constant psychrometric quality for a given inlet air condition.

3.6.1 Cabinet. The cabinet enclosure shall be constructed of steel or aluminum. The frame or chassis shall be rigid and of adequate strength to support and maintain alinement of the assembled parts. Removable panels shall be provided to permit access for removal of the desiccant bed or contractor, air seals, and heater. Fans and motors shall be installed within the cabinet or mounted externally. The removable panels on each dehumidifier shall be interchangeable. Separate openings shall be provided for access to the air filters. Gaskets shall provide weathertight seals at all access openings.

3.6.1.1 Airflow connections. Airflow connections shall be provided for a conditioned air inlet, a conditioned air outlet, a reactivation air inlet, and a reactivation air outlet. The inlets and outlets shall have smooth, flat surfaces for flanged duct connections. The dehumidifier shall be furnished with gasketed blank flanges at the air connection openings.

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3.6.1.2 Air passage connections. Connections in all air passages shall be welded or attached by bolted flanges and shall be airtight. Flanged connections shall have gaskets of a material capable of withstanding an intermittent temperature of 392°F. The flanges on inlet and outlet connections shall be constructed to minimize strain transmission from external ducts to internal components of the dehumidifier. The manufacturer shall provide adequate means to drain any condensate that may collect in the reactivation air outlet passage.

3.6.2 Desiccant bed or contractor. The desiccant bed shall contain a granular desiccant. The desiccant contactor shall contain a nongranular desiccant. The desiccant shall have adequate capacity to meet the performance requirements of 3.5.1. Free liquid shall not result during the absorption cycle, and the desiccant shall be adequate for repeated reactivation. There shall be no measurable carryover of desiccant from the bed or contactor to the conditioned air outlet at maximum inlet airflow and design dewpoint. Airflow shall be equally distributed over the entire cross-sectional area of the bed or contactor. The bed or contactor shall be so supported as to prevent self-grinding, settling, erosion, and attrition due to air friction, thermal stresses, or other conditions encountered during normal operation. The static pressure loss at design velocity through the desiccant bed or contactor shall be not greater than 2.0 inches of water. The bed or contactor shall not be capable of producing flammable or noxious gases at temperatures up to 392°F. If subjected to a temperature of 392°F, the bed or contactor shall operate at maximum efficiency when the air returns to normal design operating temperature.

3.6.2.1 Granular desiccant bed. The desiccant bed shall consist of a granular desiccant contained in corrosion-resistant or cadmium-plated, perforated metal. The metal shall have adequate strength to support the desiccant bed without deformation. The perforations shall be of a size that prevents loss of the granular desiccant. Adequate means shall be provided to replace the desiccant without disassembly of the dehumidifier. The desiccant bed shall have adequate capacity to ensure a full bed in the active zone.

3.6.2.2 Nongranular desiccant contactor. The contactor shall consist of a nongranular, fixed-desiccant impregnated structure. The desiccant shall be uniformly dispersed throughout the contactor.

3.6.2.3 Drive system. The desiccant bed or contactor shall be driven by an adjustable system consisting of a V-belt, a chain, or a combination V-belt and chain. Belt tension shall be controlled by positive action belt adjustment or shall be automatically maintained. The drive shall provide consistent performance. Lubrication, when required, shall be in accordance with 3.7.2.

3.6.2.4 Airstreams. The reactivation airstream shall flow through the desiccant bed or contactor in a counter direction to the conditioned airstream for automatic cleaning of the desiccant. The reactivation airstream shall pass directly from the heater to the desiccant structure. Dehumidifiers shall be equipped with airflow gauges and reactivation and process air circuits.

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3.6.3 Air Seals. Air seals shall be provided to prevent leakage between the conditioned air and reactivation air circuits, and to prevent air from bypassing the desiccant bed or contactor. The seals shall not crack, harden, or deteriorate when subjected to normal operating temperatures. Seals shall be long wearing and shall not cause damage to the desiccant bed or contactor. The seals shall be adjustable or self-adjusting, easily removable, and provide convenient access to facilitate inspection of components.

3.6.4 Air filters. All air shall be filtered before it enters the desiccant bed or contactor in the conditioned air circuit and prior to entering the heater in the reactivation air circuit. The filter shall be the dry, cleanable, or disposable type and shall be rigidly constructed. The filter shall consist mainly of a nonferrous or mineral substance which acts as a media. The filter shall have high arresting efficiency, high dirt-holding capacity, and be cleanable or disposable. The filters shall be reading accessible to permit cleaning and replacement.

3.6.5 Thermal insulation. Exposed dehumidifier surfaces, having a temperature of 130°F or higher during any service condition, shall be insulated to protect personnel and to prevent heat transfer to surroundings and from any component when the transfer of heat is detrimental to operation of the component or the system. Surfaces subject to condensation shall be insulated to prevent dripping or runoff of moisture when the dehumidifier is operated at rated capacity. The insulation shall be a glass fiber blanket or a sheet of formed insulation of foamed closed-cell plastic. Plastic insulation shall not be used when surfaces to be insulated have a temperature of 200°F or higher. The insulation shall be impervious to water vapor, smooth, and scuff-resistant. Loose insulation applied to outside surfaces or installed within air passages shall be protected by aluminum foil or a jacket of sheet aluminum or galvanized steel.

3.6.6 Reactivation heaters. The desiccant shall be reactivated by electric, gas, or steam heaters as required (see 1.2). The heating surface shall be installed in the reactivation inlet air passage leading to the desiccant bed or contactor. During normal operation, surface temperatures of electric heaters shall not exceed 900°F. A manual reset, limit switch shall stop operation of gas or electric heaters if the reactivation air temperature exceeds 325°F. An airflow sensing device shall be provided to assure that the required amount of air is supplied for operation of the gas heater. With an adequate airflow, operation of the sensing device shall be automatic causing a gas valve to become energized and the heater to reactivate the desiccant. No concentration of excessive heat shall result as a consequence of inadequate heating element design.

3.6.6.1 Type I (electric) heaters. Type I heaters shall be suitable for forced air heating. Unless otherwise specified (see 6.2), heaters shall operate on a nominal 208-volt, 3-phase, 60-Hertz (Hz) alternating current (ac) system. Electrical junctions shall be secured in a manner to prevent an increase in resistance and thereby cause the temperature at a junction to exceed the normal operating temperature of the heating element. The heater shall be capable of withstanding an intermittent temperature 25 percent above normal operating temperatures. The electric energy required to operate heaters and fan motors shall not exceed 20 kilowatts. Heaters shall be installed to permit inspection, servicing, and replacement without disassembly of the dehumidifier.

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3.6.6.2 Type II (gas) heaters. Type II heaters shall be direct or indirect gas-fired. The reactivation airstream shall be heated without resulting in excessive heating of the body or casing of the dehumidifier or the desiccant. Controls for gas heaters shall be provided. Unless otherwise specified (see 6.2), the controls shall include a lubricated plug shut-off cock, gas pressure regulator, motorized gas valve, firing rate cock, pilot cock, pilot regulator, pilot gas solenoid, spark-ignited pilot with ignition transformer, electronic main flame supervision, thermal limit switch, and airflow sensing device (see 3.6.6). The total energy required to operate the heater, and the burner fan, if provided, shall not exceed 83,000 British thermal units per hour (Btu/h). The burner shall be installed to facilitate inspection, servicing, and replacement without disassembly of the reactivation air duct or the dehumidifier.

3.6.6.3 Type III (steam) heaters. Type III heaters shall have nonfreeze, distributor tube-type steam coils with headers and threaded ends for connection to steam supply and condensate return. The coil section shall be enclosed in a galvanized steel or aluminum casing. The total energy required to operate the heater shall not exceed 73 lb. of steam per hour at 100 pounds per square inch (psig) gauge. The coils shall be installed to facilitate inspection, servicing, and replacement without disassembly of the dehumidifier.

3.6.7 Fans. The conditioned air fan and the reactivation air fan shall be the centrifugal type. The fans shall be direct or belt driven, statically and dynamically balanced. Fan housings shall have smooth, curved inlets. Fan wheels and housings shall be steel or aluminum. The conditioned air fan and the reactivation air fan shall have the capacity to overcome the internal resistances of the dehumidifier and to produce not less than 0.5 inch of static water pressure at each air outlet. The fan wheel shall be the multiblade type, smooth, and free from projecting bolts, nuts, and other projections that may catch dirt or lint and cause noisy operation. The conditioned air fan shall remove humid air from the space to be dehumidified and circulate the air through the desiccant bed or contactor. The reactivation air fan shall circulate outside air through the desiccant in an opposing direction to the flow of conditioned air. Adequate clearance shall be provided between stationary and moving components.

3.6.8 Electrical requirements. Each dehumidifier shall be furnished complete with conditioned air and reactivation air fan motors, necessary controls, and wiring.

3.6.8.1 Motors, motor starters, and motor overload protective devices. Motors shall be the squirrel cage induction, polyphase type, having class B insulation, dripproof enclosure, of adequate size for continuous operation without overheating, and rated and constructed in accordance with NEMA MGI. Motors shall be ballbearing, provided with a motor starter, thermal overload protection of the manual reset type, and under-voltage protection. The overload relay contacts shall be wired in series in order to stop operation of the dehumidifier when either motor becomes overloaded. The motor starters and overload protection devices shall be housed in a dripproof controller meeting the requirements of NEMA ICS2 and enclosure in accordance with NEMA ICS6 type 2. Powerlines shall have flexible connections to eliminate vibration transmission. Unless otherwise specified (see 6.2), electric motors shall operate on a nominal 208-volt, 3-phase, 60-Hz, ac system. The continuous horsepower rating of each motor shall be adequate to assure optimum performance and maximum efficiency.

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3.6.8.2 Controls. With the exception of a humidity controller and a circuit breaker, each dehumidifier shall be equipped with integrally mounted controls. A selector switch having three points of electrical contact for automatic, manual, and off positions shall be mounted in an easily accessible location on the control box. The switch shall normally be set for automatic control. The dehumidifier shall automatically start and stop operating in response to the remote humidity controller. When the switch is set for manual operation, the humidity controller shall be shunted, thereby providing continuous operation of the dehumidifier. The controls shall operate on a nominal 110-volt, single-phase, 60-Hz, ac system. The control box shall be sheet steel, and shall contain a hinged door.

3.6.8.3 Electrical wiring and components. The equipment shall be completely wired for operation, requiring only connection to the electrical source of supply. Adequate provision shall be made to ground the equipment. Wiring and electrical components, such as controllers and switches, shall conform to the applicable requirements of NFPA No. 70. All wire shall be color-coded and suitably identified at all terminals and junctions. Unless otherwise specified (see 6.2), electrical components shall be designed for operation on a nominal 208-volt, 3-phase, 60-Hz, ac system.

3.6.8.4 Circuit breaker. Each dehumidifier shall be provided with a 3-pole, quick-make, quick-break circuit breaker in accordance with the applicable requirements of NEMA AB1 and UL 489. The circuit breaker shall be mounted in the controller or in a dripproof enclosure adjacent to the control box.

3.6.8.5 Hour meter. When specified (see 6.2), the dehumidifier shall be equipped with an hour meter. The meter shall register the total operating time. The hour meter shall include a 110-volt synchronous motor connected through adequate gearing to a counter that registers not less than 9,999 hours of operating time.

3.6.8.6 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.6.8.7 Electromagnetic interference control. When specified (see 6.2), the equipment shall meet the requirements of class C3, group I of MIL-STD-461.

3.6.9 Lifting attachments. All dehumidifier units covered by this specification shall be equipped with a lifting device. The lifting device shall be designed and located to permit the units to be lifted in the normal travel position. Each lifting device shall be designed to withstand a loading equal to 2-1/2 times the normal static load imposed when the dehumidifier is suspended in a stationary position.

3.7 Maintainability. The dehumidifier shall be designed to facilitate maintenance and servicing of components in the field.

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3.7.1 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7.2 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated as specified in section 5. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, (1,000 psig or higher), will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. Pressure lubrication shall not be used for motor bearings. A lubrication chart (metal or decal) shall be affixed to each machine to indicate all points requiring lubrication, the grade of lubricant required, and the time interval for accomplishing lubrication. Although initially serviced with standard commercial lubricants, the units shall be capable of operating efficiently and satisfactorily without deleterious effects with standard military lubricants (see 6.5).

3.8 Treatment and painting. Unless otherwise specified (see 6.2), the dehumidifier shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the dehumidifier other than corrosion-resisting steel shall be protected against corrosion and present a neat appearance.

3.9 Identification marking. Identification shall be permanently and legibly marked directly on the dehumidifier or on a corrosion-resisting metal plate securely attached to the dehumidifier at the source of manufacture. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.10 Instruction plates. The dehumidifier shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.11 Spare parts. When specified (see 6.2), spare parts shall be furnished and shipped with each unit. The spare parts required and the quantity thereof shall be as specified in the contract or, when applicable, shall be determined in accordance with the provisioning procedures of the contract.

3.12 Workmanship.

3.12.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

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3.12.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.12.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes and in full contact with the surface of the member.

3.12.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.12.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. The first article inspection shall be performed on one assembly when a first article sample is required (see 3.1 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the assembly meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining assemblies to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.6. This inspection shall be performed on each unit.

4.3 Examination. Each dehumidifier shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Capacity test. The unit shall be tested to verify compliance with the capacity requirements of 3.5.1. The test shall be performed with reactivation air and conditioned air at each test condition listed in table I. The dehumidifier shall be located in a space having a constant temperature. The temperature may be within the range of 75°F to 85°F. The dehumidifier shall be operated until equilibrium has been established. Equilibrium shall be considered established when the total moisture entering the system is within ±5 percent of the total moisture exiting the system. Provision shall be made to measure and record inlet and outlet temperature and humidity, power input, rate of airflow, total airflow pressure at the outlets, temperature or air leaving the heater, and other necessary data. The data shall be obtained at 15-minute intervals for a minimum period of 1 hour after the test area has reached test conditions. Failure to meet the capacity requirements shall constitute cause for rejection.

4.4.2 Energy consumption test. The unit shall be tested to verify compliance with the energy consumption requirements of 3.5.2. The tests shall be conducted at each condition listed in table II. Failure to meet the energy consumption requirements shall constitute cause for rejection.

4.4.3 Operational tests. Each unit shall be operated to determine that the dehumidifier, safety devices, and controls are adequately constructed. The safety devices and controls shall be checked to ensure proper performance. The insulation resistance and the dielectric strength shall be measured to determine that personnel are not unduly exposed to hazardous conditions. The unit shall be operated continuously for 24 hours at a temperature within the range of 65°F to 85°F. If not in service during the test, the heaters shall be connected and the unit operated for 2 hours. Evidence of faulty construction, overheating of any parts, hazards to personnel, unsatisfactory operation of control equipment including thermal switches, relays, and other safety devices, or deficiency in capacity shall constitute cause for rejection.

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4.4.4 Air leakage test. The air leakage test shall be performed on each unit tested in 4.4.3 before the unit has cooled. The test shall be performed both with the desiccant bed or contactor in a stationary position and in motion. With the conditioned air outlet blocked, a measured quantity of air shall flow through a gas flowmeter to the conditioned air inlet. The test may be performed by the introduction of air through the reactivation air inlet while the reactivation air outlet is blocked. The static pressure differential between the conditioned air and the reactivation air circuits shall be not less than 1.0 inch of water. The rate of leakage shall not exceed 3 percent of the rated conditioned airflow.

4.4.5 Dielectric strength test. The dehumidifier and accessory equipment, with the exception of motors, shall withstand for a period of 1 minute, without breakdown, a minimum dielectric test voltage of 1,000 volts plus twice the rated voltage. All motors shall be tested to meet the applicable NEMA dielectric strength requirements for the specific type and rating. The frequency of the test voltage shall be not less than 60-Hz.

4.4.6 Electric heater tests. The insulation resistance test and the dielectric strength test shall be performed in accordance with MIL-H-22577.

4.4.7 Lifting devices test. The static load, as specified in 3.6.9, shall be applied to the attachment when mounted on its structural member. Weld failure or deformation of the attachment or structural member shall constitute cause for rejection.

4.5 Electromagnetic interference control test. When conformance to electromagnetic interference suppression limits is required, the equipment shall be tested in accordance with MIL-STD-462 to determine conformance to the electromagnetic suppression characteristics of 3.6.8.7 (see 6.2).

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation are listed in MIL-P-116, and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

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5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedures of MIL-P-116.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as in 5.1.1.5.1 and 5.1.1.5.2.

5.1.1.5.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.5.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2, or MIL-B-22191, type II. The material shall be secured in place with waterproof tape.

5.1.1.6 Electrical motors. Preserve level A by the alternate method of MIL-E-16298.

5.1.1.7 Electrical equipment. Electrical equipment shall be preserved and packaged in accordance with the level A requirements of MIL-E-17555.

5.1.1.8 Drive system.

5.1.1.8.1 Drive Chains.

5.1.1.8.1.1 Exposed drive chains. Exposed drive chains shall be coated with enough type P-3, P-9, or P-21 preservative to ensure penetration of the preservative to the inner surface of the rollers, pins, and bushings. After the excess preservative has drained, the entire chain and the unpainted surfaces of the sprocket shall be coated with type P-1 preservative.

5.1.1.8.1.2 Enclosed chains. Enclosed chains which operate in lubricants shall be preserved by having the housing filled to the operating level with one of the following applicable preservatives:

Operating Lubricant	Preservative Lubricant
Lubricating oil (SAE 10, 30, or 50)	P-10, grade as applicable
Gear lubricant (SAE 75, 80, 85, 90, or 140)	MIL-L-2105, grade as applicable
Lubricant other than above operating lubricant	Manufacturer's approved

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Chains shall be operated to ensure coating of all interior surfaces with preservative. Chain housings containing any of the above lubricants shall be identified with a weatherproof tag to indicate: "This housing is filled to the operating level with lubricant. Do not drain until first required lubrication change." Markings shall be applied to the tags with a waterproof material. The tags shall be attached in a conspicuous location.

5.1.1.8.2 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved by method IC-1 or IC-3. Unpainted surfaces shall be coated with primer conforming to TT-P-664 or MIL-P-46093. A weatherproof tag shall be attached in a conspicuous location indicating: "Belts have been (removed or released from tension). (Install or tension) prior to operation." Markings shall be applied to the tags with a waterproof material.

5.1.1.9 Desiccant. Desiccant shall be preserved by method IA with the selection of the submethod at the option of the contractor.

5.1.1.10 Lubrication. All parts requiring lubrication and not specifically provided for herein shall be lubricated in accordance with the contractor's standard practice.

5.1.1.11 Tools and repair parts. Tools and repair parts shall be preserved and packaged in accordance with the level A requirements of PPP-P-40, or when items are not specifically covered by PPP-P-40, an applicable submethod of preservation of MIL-P-116 shall be used.

5.1.1.12 Technical publications. Technical publications for each piece of equipment shall be preserved by method IC-1 or IC-3.

5.1.1.13 Consolidation. Loose components, tools, repair parts, and publications shall be consolidated in boxes conforming to PPP-B-636, class weather resistant.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2.).

5.2.1 Level A. The complete unit shall be packed in a box conforming to PPP-B-601, military overseas type; PPP-B-621, class 2; MIL-B-26195, type II with plywood superstructure, or a crate conforming to MIL-C-104, type II, style A. The gross weight shall determine the container selected. Contents shall be cushioned, blocked, braced, and anchored in accordance with MIL-STD-1186. Contents packed in boxes conforming to PPP-B-621 and crates conforming to MIL-C-104, class 1 shall be shrouded in accordance with MIL-STD-1186.

5.2.2 Level B. The complete unit shall be packed in a box conforming to PPP-B-601, domestic type; PPP-B-621, class 1; or a crate conforming to MIL-C-52950. Contents shall be cushioned, blocked, braced, and anchored in accordance with MIL-STD-1186. Contents packed in MIL-C-52950 crates shall be shrouded in accordance with the appendix to the crate specification or MIL-STD-1186.

5.2.3 Commercial. The equipment shall be prepared for shipment in a manner which will ensure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

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5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Dehumidifiers covered by this specification are intended for dehumidification of warehouses or other storage spaces.

6.2 Acquisition requirements. Acquisition documents should specify the following.

a. Title, number, and date of this specification.
b. Type of reactivation heater required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2, and 2.1.1).
d. If a first article is required for inspection and approval (see 3.1, 4.2.1, and 6.4).
e. Power requirement for electric heaters if other than specified (see 3.6.6.1).
f. Control requirements for gas heater if other than specified (see 3.6.6.2).
g. Power requirements for motors, motor starters, and motor overload protective devices if other than specified (see 3.6.8.1).
h. Power requirements for electrical components if other than specified (see 3.6.8.3).
i. If hour meter is required (see 3.6.8.5).
j. If fungus resistance is required (see 3.6.8.6).
k. If electromagnetic interference control is required (see 3.6.8.7, 4.5 and 6.3.1).
l. If means for lubrication is to be other than manufacturer's standard practice (see 3.7.2).
m. If treatment and painting are to be other than manufacturer's standard practice (see 3.8).
n. If spare parts are required (see 3.11).
o. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.3.1 Quality assurance acceptability. The following alternative method should be considered in lieu of the performance test as specified in section 4 when this specification is applied on a contract (see 6.2).

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Reference paragraph:

4.5 - When suppressed to conform to 3.6.8.7, the contractor may furnish a certificate of compliance in lieu of the test, together with a list of suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items(s) should be a first production item or a standard production item from the contractor's current inventory (see 3.2), and the number of items to be tested as specified in 4.2.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for sample for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Lubricants. If deemed necessary, procurement documents should specify that the contractor furnish a list of lubricants recommended for the dehumidifier. This list should include the type, grade, manufacturer's name, and when applicable and available, the appropriate military specification number. To assist manufacturers in identifying suitable military lubricants, the contracting officer should be prepared to supply to the contractor, when requested, a list of applicable lubricants as contained in the Federal Supply Group 91 and pertinent specifications. The finalized and approved list of lubricants should then be included in the technical publications furnished with the units.

6.6 Part or Identifying Number (PIN). The PIN corresponds to the type of dehumidifiers covered by this specification, and defines the requirements of the options presented under this specification. The military specification number and the type code number are combined to form the PIN.

6.6.1 Cataloging data. For cataloging purposes, the PINs for the dehumidifier are assigned as follows:

M 28594 - 01, 02, or 03

Prefix to indicate military specification

Specification number

Type of reactivation heater (see 6.6.2)

6.6.2 Type. The type of dehumidifier (see 1.2) is identified by a single numerical character (see table III).

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TABLE III. Code number to type.

Type	Code
I - Electric	01
II - Gas	02
III - Steam	03

6.7 Subject term (key word) listing.

Air dryer
Climate control
Dehydrator
Dryer
Moisture removal

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
Air Force - 84

User activities:
Army - CE, GL
Navy - MC

Preparing activity:
Navy - YD

(Project 4440-0055)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-28594B	DEHUMIDIFIER, ROTARY, DRY-DESICCANT TYPE AUTOMATIC, CONTINUOUS DUTY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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